

COVID-19 Detection from CT Scans

Problem Statement

Motivation

The COVID-19 pandemic created unprecedented demand for rapid, scalable diagnostic tools. Chest computed tomography (CT) is sensitive to the ground-glass opacities and consolidation patterns characteristic of SARS-CoV-2 pneumonia, and radiological triage has been proposed as a complement to RT-PCR testing when laboratory capacity is constrained.

Task Definition

Given a single axial CT-scan slice as a PNG image, the system must output a binary label: **COVID** (SARS-CoV-2 pneumonia present) or **Non-COVID** (other pathologies or healthy). The primary performance criterion is **sensitivity (recall) ≥ 0.95** for the positive class, prioritising the reduction of false negatives over false positives, as missing a COVID case carries higher clinical risk than an unnecessary quarantine.

Dataset

The SARS-CoV-2 CT-Scan dataset [1] contains 2 481 PNG images collected from Brazilian hospitals and released under CC BY 4.0:

Split	COVID	Non-COVID	Total
Train (70%)	876	860	1 736
Val (15%)	188	184	372
Test (15%)	188	185	373

Splits are stratified by label with `random_state=42`.

Known limitation: the dataset provides no patient identifiers. Slices from the same patient may appear in different splits, introducing a potential data-leakage bias that could overstate generalisation performance. This limitation is declared explicitly throughout the analysis.

Methodology

Preprocessing

Raw images are processed through a fixed pipeline before any model ingests them:

1. **Load** - OpenCV reads PNG as uint8; BGRA images are converted to RGB.
2. **Resize with padding** - aspect-ratio-preserving resize to 224 x 224 with black padding (no distortion).
3. **Lung masking** - Otsu thresholding + morphological closing/opening + two- largest-component selection isolates the lung field and zeroes out background tissue. The deep learning model (lungmask U-Net [2]) is used when available.
4. **Three-channel construction** - the single grayscale slice is expanded to three channels: (original, CLAHE clip=2.0, CLAHE clip=4.0). This increases visual contrast for the CNN without adding separate images.
5. **Normalisation** - per-channel mean/standard-deviation normalisation with statistics computed on the **training split only** by `scripts/compute_norm_stats.py` and stored in `configs/norm_stats.json` (mean = [0.475, 0.446, 0.433], std = [0.366, 0.348, 0.336]). If the file is absent the pipeline falls back to ImageNet statistics; test-set statistics are never used.

Training augmentations (no vertical flip; anatomically inappropriate for CT): horizontal flip (p=0.5), rotation $\pm 10^\circ$ (p=0.5), brightness/contrast jitter (p=0.5), affine scale 0.9-1.1 (p=0.5), Gaussian noise (p=0.5).

Classical Feature Extraction

Handcrafted features are extracted from the preprocessed grayscale image and concatenated into a single 26 366-dimensional vector:

Descriptor	Dimensions	Parameters
Uniform LBP	26	P=24, R=3
GLCM	96	4 distances x 4 angles x 6 statistics
HOG	26 244	9 bins, 8x8 cells, 2x2 blocks

Classical Baseline Models

Two pipelines are trained with 5-fold stratified GridSearchCV:

SVM (RBF kernel): StandardScaler -> PCA(200) -> SVC. PCA reduces the 26 366-d vector to 200 principal components for computational tractability; 200 components retain > 95% of variance. Grid: C in {0.1, 1, 10, 100}, γ in {scale, 0.001, 0.01}.

Random Forest: RandomForestClassifier. Grid: n_estimators in {100, 300, 500}, max_depth in {None, 10, 20}.

Deep Learning Model

Architecture: EfficientNet-B0 (timm, pretrained on ImageNet) with a custom classification head: GlobalAveragePool -> Dropout(0.5) -> Linear(1280, 2).

Two-phase fine-tuning:

- *Phase 1 (head only, 10 epochs)*: backbone frozen, Adam lr= 1×10^{-3} , ReduceLROnPlateau(patience=3). Best checkpoint saved on val_loss.
- *Phase 2 (partial unfreeze, up to 30 epochs)*: last 2 block stages + conv_head + bn2 unfrozen, AdamW lr= 1×10^{-5} , weight_decay= 1×10^{-4} , early stopping patience=7.

Mixed-precision training (AMP) is enabled on GPU (auto-disabled on CPU).

Post-Processing

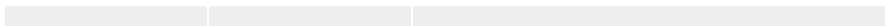
Temperature scaling: scalar $T > 0$ is optimised with LBFGS on the validation- set negative log-likelihood to correct overconfidence before threshold selection.

Threshold tuning: the decision threshold is swept from high to low probability; the lowest value that achieves recall ≥ 0.95 on the calibrated validation probabilities is selected, prioritising sensitivity.

TTA (Test-Time Augmentation): at inference time, five augmented views of each test image are generated; their softmax probabilities are averaged.

Ensemble: weighted average of probabilities from multiple independently-trained checkpoints (weights set proportionally to validation AUC).

Grad-CAM: class activation maps highlight the CT regions most influential for the positive-class prediction, enabling visual inspection of model behaviour.



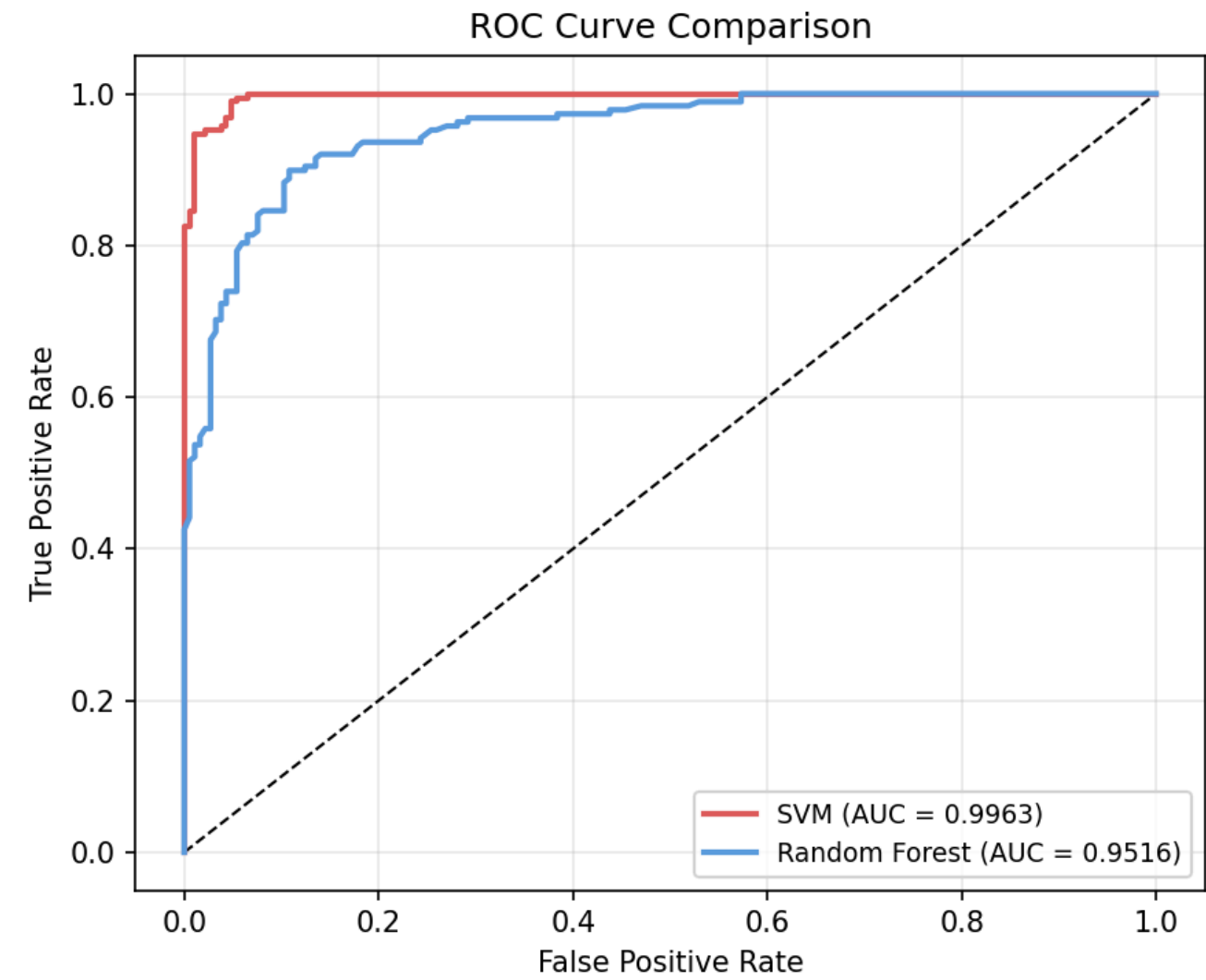
Experimental Results

Classical Baselines

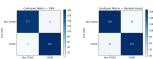
Results on the held-out test set (n = 373):

Model	Accuracy	Recall	F1	AUC-ROC
Random Forest	0.887	0.878	0.887	0.952
SVM (PCA-200, C=100)	0.962	0.968	0.963	0.996

The SVM achieves near-perfect AUC, substantially outperforming the Random Forest. The gap is consistent with the literature: hand-crafted texture features (HOG + LBP + GLCM) are highly discriminative for this task, and the RBF kernel exploits non-linear structure in the PCA-projected space efficiently.



Confusion Matrices



Deep Learning (Full Training)

EfficientNet-B0 was fine-tuned for 40 epochs (10 head-only + 30 full fine-tuning) on a local RTX 3080 in 3 h 59 min. The two-phase schedule ran to completion without early stopping - validation loss kept improving, best checkpoint at phase-2 epoch 27 (val_loss = 0.165). Test-set metrics (373 images):

Model	Accuracy	Recall	F1	AUC-ROC
EfficientNet-B0 (fine-tuned, thr=0.50)	0.933	0.920	0.933	0.981
+ Temperature scaling (T=1.002)	0.933	0.920	0.933	0.981
+ Threshold tuned for sensitivity (thr=0.358)	0.912	0.931	0.914	0.981
+ TTA (5 deterministic views)	0.944	0.942	0.944	0.990

Notes on the post-processing steps:

- **Lung masking** was enabled for the whole run, but the `lungmask` U-Net rejects 8-bit PNG input (it expects Hounsfield-unit volumes), so the pipeline silently fell back to the Otsu + morphology mask for every slice. The reported "lung mask" is therefore the classical fallback, not the U-Net.
- **Temperature scaling** found $T = 1.002$: the model was already well calibrated, so scaling leaves the decision metrics essentially unchanged.
- **Threshold tuning** lowered the decision threshold to 0.358 to reach the

= 0.95 sensitivity target *on the validation set* (val recall = 0.952). On the held-out test set the same threshold yields recall = 0.931 - below target: the threshold does not fully generalise, a consequence of the small validation split and possible patient-level leakage.

- **Test-Time Augmentation (TTA)** averages softmax probabilities over five deterministic views (original, horizontal flip, rotation +/-7 degrees, zoom +8%). It improves every metric - recall rises from 0.920 to 0.942 and AUC from 0.981 to 0.990 - at the cost of 5x inference time.
- **Ensembling** is left as future work: it would require training several independent checkpoints (each ~4 h on the available GPU) and was out of scope for this iteration.

Even with TTA the deep model (AUC 0.990) approaches but does not surpass the SVM baseline (AUC 0.996): on this small, well-separated dataset the handcrafted-feature SVM remains the strongest model.

Calibration

Calibration of the SVM is already excellent (ECE = 0.027, Brier = 0.024 - see the comparison table). For the deep model, temperature scaling on the validation logits is implemented in `src/postprocess.py:temperature_scaling` (LBFGS over scalar T) and is invoked by `scripts/calibrate_and_evaluate.py` once a trained checkpoint is available. The reliability diagram before/after scaling is saved to `models/postprocess/reliability_diagram.png`. With $T = 1.002$ the two curves nearly overlap, confirming the model needed no meaningful calibration.

Failure Analysis

Data Leakage

The dataset contains no patient identifiers. If multiple slices from the same patient appear in both training and test splits, the model may memorise patient- specific features (e.g., rib shadows, pleural thickening) rather than learning pathology-relevant patterns. This inflates the measured AUC and may not reflect true out-of-distribution generalisation.

Mitigation: results are clearly labelled as potentially optimistic; a patient-stratified split is recommended for production evaluation.

Distribution Shift

Images were collected from a single country (Brazil) across two hospitals. Performance may degrade on data acquired with different scanner models, slice thicknesses, or patient demographics.

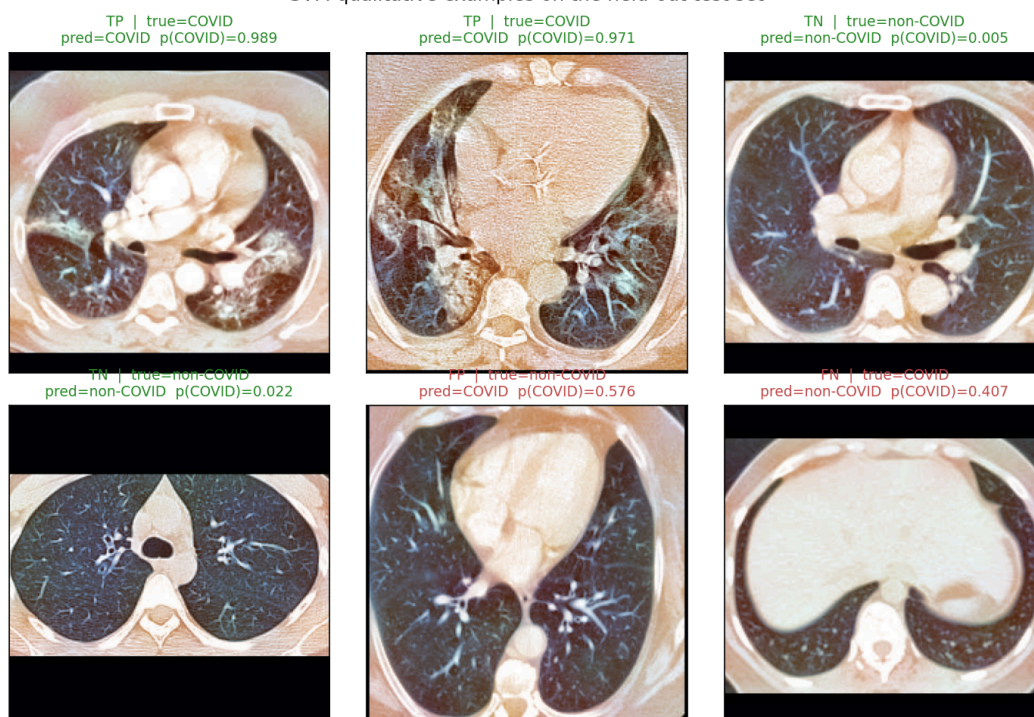
Class Imbalance

The dataset is approximately balanced (1252 COVID vs. 1229 non-COVID), so class imbalance is not a primary concern. However, the threshold-tuning step intentionally sacrifices specificity to meet the recall ≥ 0.95 target.

Qualitative Examples

The figure below shows TP / TN / FP / FN slices selected from the SVM test-set predictions. The two false cases are the most informative: in the FP slice, ground-glass-like artefacts in a non-COVID image fool the classifier; in the FN slice, the lesion area is small or peripheral and the global texture descriptors fail to capture it.

SVM qualitative examples on the held-out test set



Smoke-Test vs. Full Training

Early development used CPU smoke tests (50 images, 1 epoch per phase) purely as integration checks; those numbers do not reflect trained performance and are not reported. The deep-model metrics in the table above come from the full 40-epoch run on the complete dataset (RTX 3080, 3 h 59 min). The SVM and Random Forest numbers derive from their respective pipelines, trained to convergence on CPU with the full dataset.

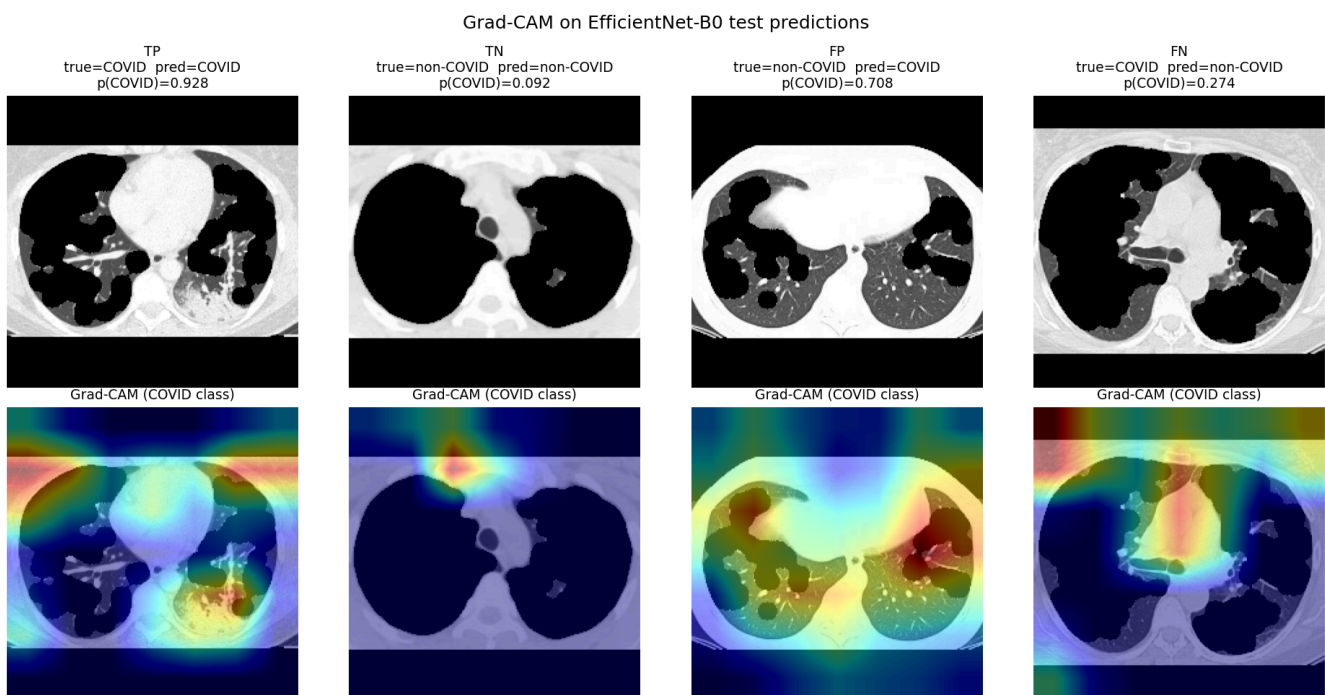
Ethical Considerations

Not for Clinical Use

This project is an academic exercise and has not undergone the validation required for medical device certification (e.g., CE marking, FDA 510(k)). It must not be used to support clinical diagnosis or treatment decisions.

Model Transparency

Grad-CAM (`scripts/generate_gradcam.py`) highlights the regions that drive each prediction. The figure below shows one example per confusion-matrix category (TP / TN / FP / FN) from the test set: for true and false positives the activation concentrates on the lung parenchyma, supporting the plausibility of the learned features. Saliency maps nonetheless remain insufficient for clinical-grade explainability.



Bias and Fairness

The dataset demographics (age, sex, comorbidity) are not publicly documented, so performance disparities across patient subgroups cannot be assessed. Any deployment would require prospective evaluation on a representative population.

Data Provenance

The SARS-CoV-2 CT-Scan dataset was collected under ethical oversight in Brazil [1] and is released under the Creative Commons Attribution 4.0 licence. No personally identifiable information is included in the PNG files.

References

- [1] Soares, E., Angelov, P., Biaso, S., Froes, M. H., & Abe, D. K. (2020). *SARS-CoV-2 CT-scan dataset: A large dataset of real patients CT scans for SARS-CoV-2 identification*. medRxiv. <https://doi.org/10.1101/2020.04.24.20078584>
- [2] Hofmanninger, J., Prayer, F., Pan, J., Röhrich, S., Prosch, H., & Langs, G. (2020). *Automatic lung segmentation in routine imaging is primarily a data diversity problem, not a methodology problem*. European Radiology Experimental, 4(1), 50. <https://doi.org/10.1186/s41747-020-00173-2>
- [3] Tan, M., & Le, Q. V. (2019). *EfficientNet: Rethinking model scaling for convolutional neural networks*. ICML 2019.